

Experiment Plan: Static Letter Recognition (Pilot)

Objective: Test whether calibrated flex sensor data, combined with engineered relative-position features, can reliably distinguish visually similar closed-fist ASL letters (A, E, M, N, S, T) that are hard to separate under simple bend thresholds.

Hypothesis: Relative-difference features between fingers (e.g., thumb vs. index bend) will separate the confusable letter set more reliably than raw per-finger values alone.

Letter Set: A, E, M, N, S, T (closed-fist family — chosen as the hardest case; open-hand letters are excluded from this pilot as they are already well separated)

Participants: 2–3 team members with different hand sizes, to test whether calibration normalization generalizes across users

Data Collection:

- 10 repetitions per letter, per participant
- At least 2 sessions per participant (recalibrating between sessions) to account for calibration drift
- Total pilot dataset: ~120–180 rows

Data Logged per Repetition:

Field	Description
timestamp	time of capture
session_id	groups reps under one calibration run
participant_id	anonymized ID per team member
letter	intended label
rep	repetition number
raw_f1–f5	raw ADC readings per finger
pct_f1–f5	calibrated % bent per finger
notes	flags (e.g. "glove slipped", "recalibrated")

Data Logged per Session (once):

Field	Description
session_id	matches reps above
open_f1–f5	raw ADC at fully open hand
closed_f1–f5	raw ADC at fully closed fist

Features Extracted (from pct_f1–f5):

- 5 raw calibrated bend values
- 4 thumb-relative differences (thumb – index/middle/ring/pinky)
- 3 adjacent-finger differences (index–middle, middle–ring, ring–pinky)
- 2 aggregate stats (mean bend, bend range)

→ 14 features per sample

Model Conditions Compared:

1. Current rule-based binary threshold logic (baseline)
2. Random Forest on raw 5 features only
3. Random Forest on full 14-feature set

Evaluation Method: Stratified 5-fold cross-validation (due to small sample size); accuracy + confusion matrix per condition

Success Criteria: Condition 3 shows measurably higher accuracy than Conditions 1–2, particularly on known confusion pairs (M/N, N/T); persistent errors in Condition 3 indicate a hardware-level limitation rather than a feature/model limitation.